CardioCloud

From ECG model to traceable cloud workflow.

An end-to-end biomedical AI prototype, built and validated on AWS.

Explore the case study →

MY ROLE / AI + CLOUD ENGINEERING

Trained the model · Defined AWS IaC

Integrated + validated the workflow

BIOMEDICAL AI / AWS CLOUD ENGINEERING / MLOPS



Actual UI · Exposure adjusted for presentation

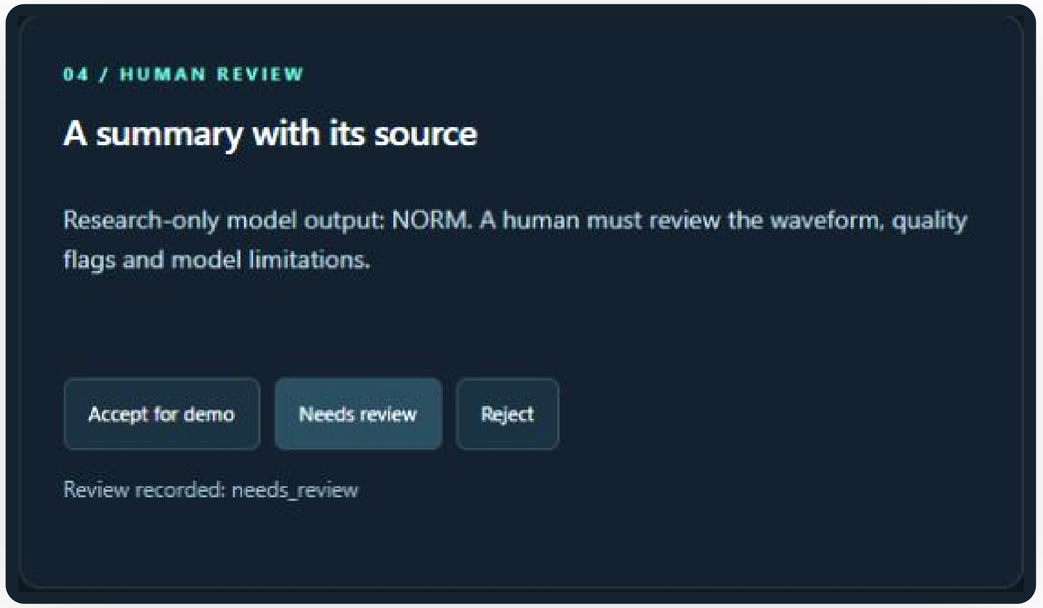
From signal to review.

A public record. A traceable run. A human in the loop.

Select an ECG → Run inference → Review the result







REVIEW CONTROLS / RECORDED STATE



Actual UI crops · Uniform exposure adjustment · No values or states altered

Measured on held-out data.

0.918

Macro AUROC

Independent PTB-XL test set

2,158 records



Public ECG data
PTB-XL v1.0.3

PyTorch model
Patient-safe partitions

Versioned artifact
Reproducible checkpoint

AWS inference
SageMaker CPU

Source: report.json / independent test split · AUROC is discrimination, not clinical accuracy

A system, end to end.

Identity, bounded processing, inference and a review that persists.

Frontend & Identity

Web + sign-in

- CloudFront
- Cognito / PKCE
- API Gateway

Private Processing

Bounded jobs

- Lambda
- Step Functions

ML & Generative AI

Model + summary

- SageMaker CPU
- Bedrock + checks

Persistence & Analytics

Review + retain

- DynamoDB
- S3 / Parquet
- Athena

Human review stays in the loop.

Private processing via service endpoints. No Internet default route in the private subnets.

Simplified view · The detailed network and IAM architecture is included as a separate resource

Proof across the lifecycle.

Real infrastructure. Recorded outcomes.

AWS deployed

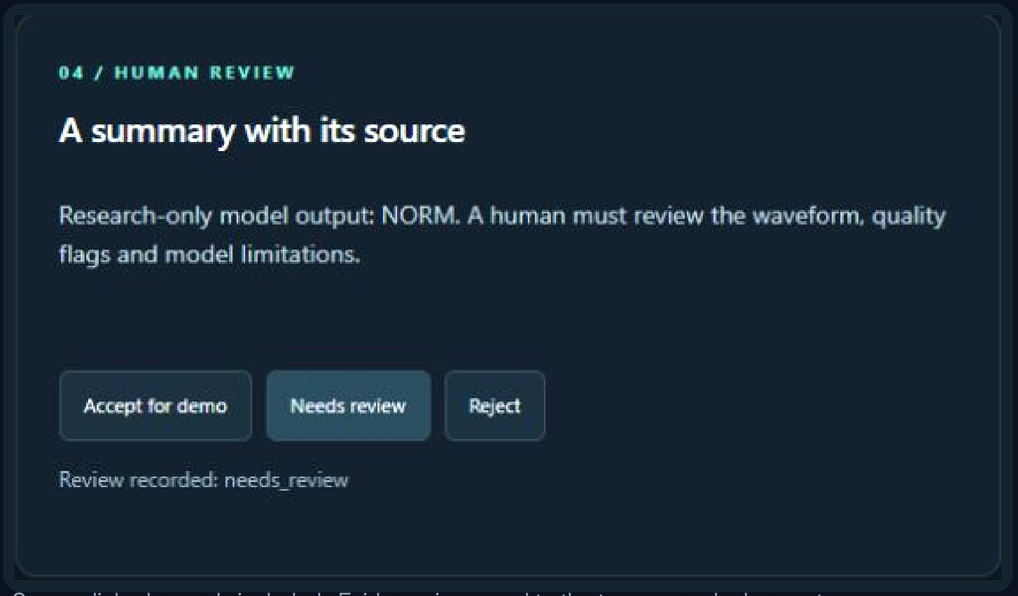
Validated in eu-west-3

80 checks

Local verification passed

6/6 cleared

Scoped post-teardown audit



Source-linked records included · Evidence is scoped to the temporary deployment

VERSIONED DELIVERY

Pinned artifact.
Recorded review.

Reproducible CPU inference and an auditable human review state.

BUILD

CodeBuild

ECR digest

SageMaker

Athena SQL

Built.
Validated.
Safely decommissioned.

Evidence retained. Infrastructure retired.

ENGINEERING DECISION / VALIDATE BEFORE DISPLAY

Strict Bedrock checks

Reject invalid generated output

Deterministic fallback

Keep the result explicit and traceable

Required human review

Require review and persist the decision

Final cleanup included scoped recovery after the original teardown failed.



Model training	AWS deployment	Validation	Evidence collection	Teardown audit
27 OCT 2026	27 OCT 2026	27 OCT 2026	27 OCT 2026	29 OCT 2026

Research prototype. Not intended for clinical diagnosis.